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1 Introduction

1.1 Purpose of the Manual

This user manual serves as a comprehensive guide for users of CodeClash Arena, a gamified competitive programming platform designed to transform traditional coding practice into an interactive and engaging experience. The manual provides detailed instructions on how to use the platform effectively, from account creation to participating in competitive battles and clan wars.

This document is intended for:

- Beginner programmers seeking to improve their algorithmic skills through competitive practice
- CSE students looking for engaging coding challenges
- Competitive programming enthusiasts wanting real-time head-to-head battles
- Coding instructors and educators implementing gamified learning environments
- Clan leaders organizing team-based coding competitions

The manual covers all essential features of the platform, including step-by-step instructions, visual guidance through screenshots, troubleshooting tips, and best practices for optimal user experience. Whether you are a first-time user or an experienced programmer, this guide will help you navigate the platform efficiently and make the most of its features.

1.2 Brief Overview of the Web Application

CodeClash Arena is a web-based competitive programming platform that combines the excitement of multiplayer gaming with professional coding contests. Inspired by popular competitive games like Clash Royale and Clash of Clans, and integrated with the Codeforces problem repository, the platform offers a unique blend of entertainment and skill development.

1.2.1 Core Concept

Unlike traditional coding platforms that focus solely on individual practice, CodeClash Arena introduces real-time competitive elements including:

- **1v1 Coding Battles**: Challenge other coders in head-to-head real-time competitions
- **Clan System**: Form or join teams and participate in clan-based battles
- **Rating System**: Track your progress through an ELO-based ranking mechanism
- **Practice Gym**: Solo practice mode with access to thousands of programming problems
- **Matchmaking**: Automatic pairing with opponents of similar skill levels

1.2.2 Key Differentiators

CodeClash Arena addresses limitations of conventional coding platforms by providing:

- Direct head-to-head competition with other coders in real-time
- Clan-based team competitions fostering collaboration and community

- Personalized contest creation with custom difficulty and time settings
- Gamified progression system with levels, XP, and visual achievements
- Real-time battle synchronization showing opponent progress
- AI-assisted learning through logical hints during practice sessions

1.2.3 Technical Foundation

The platform is built using modern web technologies:

- **Frontend**: React 19.2.0 with Vite 7.2.4 for fast, responsive user interface
- **Backend**: Node.js with Express 4.18.2 for server-side logic
- **Database**: Supabase (PostgreSQL) for data management
- **Problem Integration**: Codeforces API for accessing thousands of programming problems
- **Code Execution**: Piston API for local testing and Codeforces judge for official verdicts

1.2.4 Platform Features Summary

Feature	Description
Global Battle Mode	Join a matchmaking queue and compete against players of similar rating in real-time 1v1 battles
Local Battle Mode	Challenge specific friends directly with customizable problem difficulty and time limits
Clan Wars	Form teams, compete in clan-based tournaments, and track team performance
Practice Gym	Access solo practice mode with filtering by difficulty, tags, and rating range
Rating System	ELO-based skill rating that adapts based on opponent strength and battle outcomes
Progression System	Earn experience points and level up through battles and problem solving
Leaderboards	Global and clan rankings updated in real-time showing top performers
Battle History	Comprehensive records of past matches, opponents, and performance metrics
Code Editor	Integrated syntax-highlighted editor supporting C++, Java, Python, JavaScript, and more
AI Hints	Logical approach guidance during practice sessions without revealing solutions

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Feature	Description
Social Features	Friend lists, clan chat, mail system, and player profile viewing

Table 1.2: Core features available in CodeClash Arena

2 System Requirements

This section outlines the technical prerequisites and environmental requirements necessary to access and use CodeClash Arena effectively. Ensuring your system meets these specifications will provide optimal performance and user experience.

2.1 Supported Browsers and Devices

2.1.1 Web Browsers

CodeClash Arena is a web-based application accessible through modern web browsers. For the best experience, we recommend using the latest stable versions of the following browsers:

Browser	Minimum Version	Recommended Features
Google Chrome	Version 84+	Best performance with V8 JavaScript engine, full CSS Grid and Flexbox gap support, excellent developer tools
Mozilla Firefox	Version 103+	Strong privacy features, backdrop-filter support, reliable WebSocket connections, modern CSS features
Microsoft Edge	Version 84+	Chromium-based engine, native Windows integration, excellent performance, full modern CSS support
Safari	Version 15+	Optimized for macOS/iOS devices, complete modern CSS/JS feature support including aspect-ratio and flex gap
Opera	Version 84+	Chromium-based, built-in VPN support, efficient resource usage, modern web standards compliance
Brave	Version 1.42+	Privacy-focused, Chromium-based, full ES2020+ support, modern CSS Grid and Flexbox features

Table 2.1: Supported browsers and minimum version requirements

Important Browser Requirements:

- **JavaScript**: Must be enabled (required for React 19.2.0 and ES2020+ features)
- **Cookies**: Must be enabled for session management and authentication
- **Local Storage**: Browser must support HTML5 local storage (for user preferences and cached data)
- **WebSocket Support**: Required for real-time battle synchronization
- **Iframe Support**: Necessary for displaying Codeforces problem statements
- **Modern CSS Support**: CSS Grid, Flexbox with gap property, CSS custom properties (variables)
- **Advanced CSS Features**: backdrop-filter (with -webkit- prefix for Safari), CSS transforms and transitions
- **Modern JavaScript**: ES2020+ features including optional chaining (?.), nullish coalescing (??), async/await
- **Pop-up Blocker**: Should allow pop-ups from CodeClash Arena domain for certain features

Not Recommended or Not Supported:

- Internet Explorer (any version) - not supported, lacks modern JavaScript ES6+ features, CSS Grid, and CSS variables
- Safari versions below 15 - missing critical features like aspect-ratio, full flex gap support, and modern JavaScript operators
- Firefox versions below 103 - missing backdrop-filter support resulting in visual degradation
- Chrome/Edge versions below 84 - missing flex gap property causing layout issues
- Outdated browser versions - may cause compatibility issues, security vulnerabilities, and degraded performance
- Browsers with JavaScript disabled - critical functionality will not work

2.1.2 Device Compatibility

CodeClash Arena features a responsive design that adapts to various screen sizes and device types:

Device Type	Minimum OS/Browser	User Experience Notes
Desktop Computer	Windows 10+, macOS 11+, Linux	**Optimal Experience** - Full feature access (1920x1080+ recommended), dual-pane code editor, comfortable coding environment, all modern CSS/JS features supported, best for competitive battles
Laptop	Windows 10+, macOS 11+, Linux	**Excellent Experience** - All features accessible (1366x768 minimum), suitable for coding and battles, portable competitive environment, modern browser support required
iPad	iOS 15+ with Safari 15+	**Good Experience** - 10-inch or larger recommended, touch-optimized interface, suitable for reading problems and light coding, external keyboard strongly recommended for battles, full modern CSS support
Android Tablet	Android 10.0+ with Chrome 84+	**Good Experience** - 10-inch or larger recommended, touch-optimized interface, suitable for practice mode, external keyboard recommended, may require Chrome browser installation
iPhone	iOS 15+ with Safari 15+	**Limited Experience** - Best for viewing profiles, leaderboards, and battle history; coding on mobile not recommended for competitive battles; requires modern Safari for full feature support
Android Phone	Android 10.0+ with Chrome 84+	**Limited Experience** - Best for browsing and monitoring; on-screen keyboard not suitable for competitive coding; install Chrome browser for best compatibility

Table 2.2: Device compatibility and expected user experience

Important Device Notes:

- **Older Tablets**: iPads running iOS 14 or Android tablets below version 10.0 may experience visual issues (missing backdrop blur effects, layout problems with flex gap)
- **Browser Requirements**: Tablets must use Chrome 84+ (Android) or Safari 15+ (iOS) for full functionality
- **Tablet Mode Windows**: Surface devices and 2-in-1 laptops work best in desktop mode with physical keyboard attached
- **Mobile Limitations**: While responsive, mobile phones are not recommended for coding battles due to screen size constraints and on-screen keyboard limitations
- **Older Devices**: Devices more than 5 years old may struggle with modern JavaScript execution and CSS rendering

Screen Resolution Recommendations:

- **Minimum**: 1024x768 pixels (basic functionality, some layout compromises on smaller screens)
- **Recommended**: 1920x1080 pixels or higher (optimal experience with full CSS Grid layouts)
- **Competitive Battle**: 1440x900 pixels minimum (comfortable split-screen view for problem and editor)
- **Tablets**: 1280x800 or higher for iPad/Android tablets (portrait mode not recommended for battles)
- **Mobile**: 375x667 minimum for basic browsing (coding features disabled on small screens)

Input Device Recommendations:

- **Physical Keyboard**: Mandatory for competitive battles, required for efficient code input during timed contests
- **Mouse/Trackpad**: Strongly recommended for navigation, code selection, and editor interaction
- **Touch Screen**: Supported for browsing and navigation but not optimal for code editing; on-screen keyboards significantly slow down coding speed
- **Bluetooth Keyboard**: Acceptable for tablets, must support standard keyboard shortcuts (Ctrl+C, Ctrl+V, Tab, etc.)
- **External Monitor**: Optional but beneficial for multi-screen setup (problem viewing on one screen, coding on another)

Tablet-Specific Requirements:

- **iPad Users**: Requires iOS 15+ with Safari 15+, external keyboard mandatory for battles, landscape orientation recommended
- **Android Tablet Users**: Requires Android 10.0+ with Chrome 84+ installed, external keyboard strongly recommended

- **Stylus Support**: Not utilized by the platform, keyboard input is primary interaction method

2.2 Internet Connection Requirements

A stable internet connection is essential for CodeClash Arena as it operates entirely online and requires continuous connectivity for real-time features.

2.2.1 Minimum Internet Speed

Activity	Minimum Speed	Description
General Browsing	1 Mbps	Basic navigation, viewing profiles, accessing leaderboards
Practice Mode	2 Mbps	Loading problems, submitting code, receiving verdicts
1v1 Battles	5 Mbps	Real-time synchronization, live updates, timer synchronization
Clan Wars	5-10 Mbps	Multiple participants, real-time team coordination
Recommended	**10 Mbps+**	Optimal performance for all features, minimal latency

Table 2.3: Minimum and recommended internet speeds by activity

2.2.2 Connection Stability

- **Latency**: Less than 200ms ping for optimal real-time battle experience
- **Packet Loss**: Less than 1% for stable WebSocket connections
- **Connection Type**:
  - **Preferred**: Wired Ethernet connection (most stable)
  - **Acceptable**: Wi-Fi with strong signal strength (3+ bars)
  - **Not Recommended**: Mobile data/cellular (may cause disconnections during battles)

2.2.3 Critical Connectivity Features

- **Real-Time Synchronization**: During battles, the system uses WebSocket connections to synchronize:
  - Opponent submission status
  - Battle timer countdown
  - Live score updates
  - Problem acceptance notifications

- **External Service Access**: Your network must allow connections to:
  - Supabase servers (database and authentication)
  - Codeforces API endpoints (problem fetching and submission)
  - Piston API (code execution for practice mode)
  - Vercel hosting (frontend delivery)
- **Firewall Considerations**:
  - WebSocket connections (WSS protocol) must not be blocked
  - HTTPS (port 443) must be accessible
  - RESTful API calls must be permitted

Important Note: Unstable or slow internet connections may result in:

- Battle disconnections (automatic forfeit if reconnection fails)
- Delayed submission verdicts
- Real-time updates lag
- Timeout errors during code submission

2.3 Required Accounts and Permissions

2.3.1 CodeClash Arena Account

Required for: Access to all platform features

- **Registration Information Needed**:
  - Valid email address (for account verification and password recovery)
  - Username (unique identifier, 3-20 characters, alphanumeric)
  - Strong password (minimum 8 characters, recommended: uppercase, lowercase, numbers, special characters)
- **Account Verification**:
  - Email verification required upon registration
  - Verification link expires in 24 hours
  - Unverified accounts have limited functionality
- **Account Permissions**:
  - Read access to public leaderboards and profiles
  - Write access to personal profile information
  - Participation rights in battles, practice sessions, and clan activities
  - Submission rights for code solutions

2.3.2 Codeforces Account

Required for: Code submission and official verdict retrieval

Requirement Level: Mandatory for competitive battles and official problem solving

- **Why Codeforces Account is Needed**:
  - CodeClash Arena integrates with Codeforces for problem sourcing
  - All code submissions are forwarded to Codeforces judge for verification
  - Official Codeforces verdicts (Accepted, Wrong Answer, TLE, etc.) are retrieved
  - Problems are displayed via iframe from Codeforces platform
- **Creating a Codeforces Account**:
  - Visit `https://codeforces.com/register`
  - Complete registration with email verification
  - Choose a unique handle (username)
  - Activate account through confirmation email
- **Linking Codeforces Account to CodeClash Arena**:
  - Navigate to Profile Settings in CodeClash Arena
  - Enter your Codeforces handle in the designated field
  - System validates the handle existence
  - Handle is stored and used for all future submissions
- **Account Permissions Required**:
  - Active Codeforces account (not banned or restricted)
  - Ability to submit solutions to Codeforces problems
  - Access to Codeforces problem archive

Important Notes:

- Without a linked Codeforces account, users cannot:
  - Submit code during battles (automatic forfeit)
  - Participate in competitive ranked matches
  - Receive official verdict status
  - Contribute to clan battle scores
- Users can still access:
  - Platform browsing and exploration
  - Leaderboard viewing
  - Profile viewing
  - Problem browsing (without submission)

- Codeforces handle cannot be changed frequently (update limit implemented to prevent abuse)
- Ensure your Codeforces account is in good standing (not restricted for rule violations)

2.3.3 Additional Permissions

- **Browser Permissions**:
  - Cookies: Required for session management and authentication persistence
  - Local Storage: Required for caching user preferences and temporary data
  - Clipboard Access: Optional, improves code copying experience
- **Third-Party Service Access**:
  - Supabase (authentication and database): No additional account needed
  - Piston API (code execution): No additional account needed
  - Vercel hosting: No user account required

2.4 Software Requirements

2.4.1 Operating System Compatibility

CodeClash Arena is a cross-platform web application compatible with any operating system that supports modern web browsers:

- **Windows**: Windows 10 or later (Windows 11 recommended)
- **macOS**: macOS 11 (Big Sur) or later (required for Safari 15+)
- **Linux**: Any modern distribution (Ubuntu 20.04+, Fedora 33+, etc.)
- **Chrome OS**: Latest stable version (Chrome 84+ equivalent)
- **iOS**: iOS 15 or later (required for Safari 15+ with full feature support)
- **Android**: Android 10.0 or later (for Chrome 84+ support)

2.4.2 No Installation Required

- CodeClash Arena is entirely browser-based
- No desktop application or mobile app download necessary
- No system administrative privileges required
- Access the platform directly at: `[your-deployment-url]`

2.4.3 Optional Enhancements

- **Code Editor Extensions**: While the platform provides an integrated code editor, advanced users may prefer:
  - Copying code to external IDEs (VS Code, Sublime Text) for complex solutions
  - Using browser extensions for enhanced syntax highlighting

- **Screen Management**: For optimal battle experience:
  - Multiple monitor setup (problem on one screen, editor on another)
  - Split-screen window management
  - Browser tab management for simultaneous problem viewing

2.5 Hardware Recommendations

Component	Minimum	Recommended	Purpose
Processor	Dual-core 2.0 GHz	Quad-core 2.5 GHz+	Fast page rendering, smooth real-time updates
RAM	4 GB	8 GB or more	Multiple browser tabs, smooth multitasking
Storage	N/A (web-based)	N/A	No local installation required
Display	1024x768 resolution	1920x1080 or higher	Comfortable coding environment, split-screen view

Table 2.4: Recommended hardware configuration for smooth platform usage

2.6 Network Security Considerations

- **Corporate Networks**: Some corporate firewalls may block:
  - WebSocket connections (required for real-time battles)
  - External API calls (Codeforces, Piston API)
  - Iframe embedding (problem display)

  Solution: Contact your network administrator to whitelist necessary domains

- **VPN Users**: VPN connections may introduce latency affecting real-time battle performance
- **Proxy Servers**: Ensure proxy configuration allows WebSocket (WSS) connections

2.7 Accessibility Requirements

- **Visual**: Minimum 768px width viewport for usable interface
- **Color Scheme**: System adapts to different lighting conditions; custom themes may be implemented in future versions
- **Font Rendering**: Browser must support standard web fonts for proper code editor display

2.8 Modern Web Features and Browser Compatibility

CodeClash Arena utilizes cutting-edge web technologies that require modern browser support. The following features are implemented:

- **CSS Features**: Grid Layout, Flexbox with gap property, CSS Custom Properties (variables), backdrop-filter for blur effects, transforms and transitions
- **JavaScript Features**: ES2020+ syntax including optional chaining, nullish coalescing, async/await, arrow functions, destructuring, template literals
- **Web APIs**: WebSocket (real-time communication), Fetch API (HTTP requests), Local Storage API, History API (routing), Clipboard API
- **React 19.2.0**: Modern Hooks (useState, useEffect, useContext), latest rendering optimizations

Minimum Browser Versions for Full Compatibility:

- Chrome/Edge 84+ (for Flexbox gap support)
- Firefox 103+ (for backdrop-filter support)
- Safari 15+ (for complete modern CSS/JS features)
- Mobile Safari iOS 15+ (for full mobile compatibility)

2.9 System Requirements Summary

Requirement Category	Status
Modern Web Browser (Chrome 84+, Firefox 103+, Edge 84+, Safari 15+)	MANDATORY
JavaScript Enabled (ES2020+ support)	MANDATORY
Modern CSS Support (Grid, Flexbox gap, backdrop-filter)	MANDATORY
Internet Connection (5+ Mbps recommended)	MANDATORY
CodeClash Arena Account	MANDATORY
Codeforces Account (linked handle)	MANDATORY
Desktop/Laptop Device (for competitive battles)	MANDATORY
Physical Keyboard (for competitive battles)	MANDATORY
Stable Internet (Wired connection preferred)	**RECOMMENDED**

Table 2.5: Summary of mandatory and recommended system requirements

Note: Mobile devices (tablets, phones) can be used for browsing profiles, viewing leaderboards, and monitoring battles, but Desktop/Laptop with Physical Keyboard is mandatory for participating in competitive coding battles.

3 Getting Started

3.1 Accessing the Platform

- Open a supported web browser (Chrome 84+, Firefox 103+, Edge 84+, Safari 15+)
- Navigate to the CodeClash Arena platform
- Ensure JavaScript is enabled
- The homepage loads with title "HELLO CODERS, WELCOME!"
- On the homepage, you will see: **GET STARTED** button, **About us** link, and **Learn More** link

3.2 Sign-Up Process

3.2.1 Step 1: Navigate to Sign-Up Page

- Click **"GET STARTED"** on the homepage
- Click **"Sign Up"** link at the bottom of the Login page

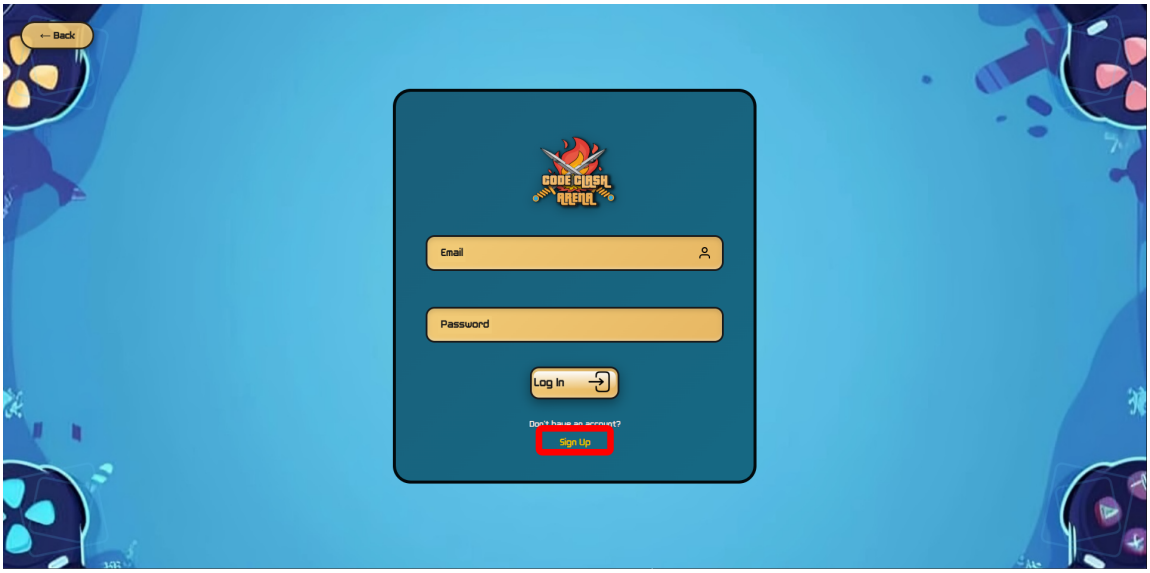


Figure 3.1: CodeClash Arena Loginpage - Locate Sign Up button

3.2.2 Step 2: Enter Registration Information

- **Email** - For verification and password recovery; must be unique
- **Codeforces Handle** - Your existing Codeforces username (create at `https://codeforces.com/register` if needed)
- **Password** - Minimum 8 characters; keep separate from Codeforces password

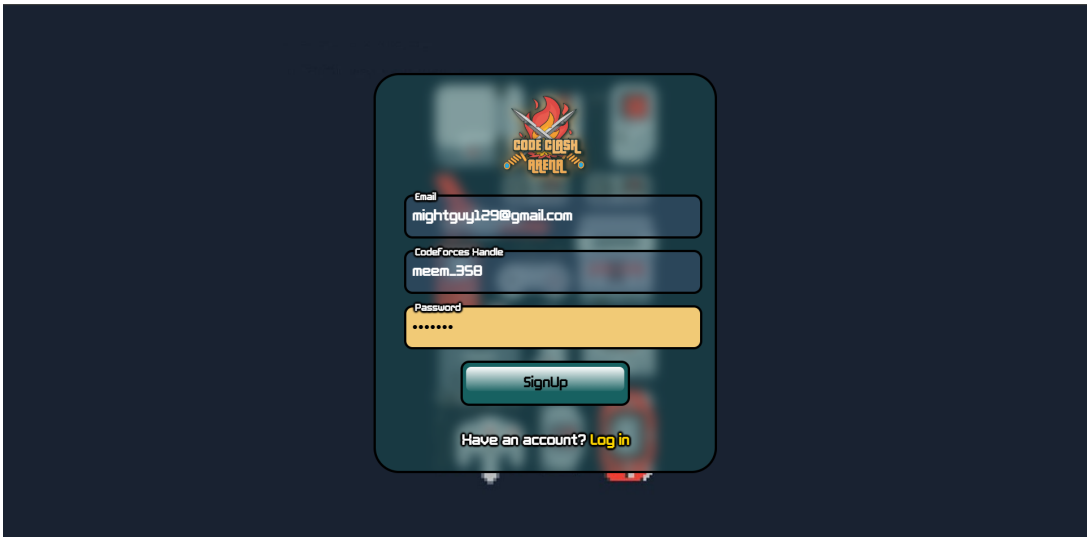


Figure 3.2: Sign-Up Form - Enter registration details

3.2.3 Step 3: Submit Registration

- Click **"SignUp"** button
- If validation fails, red field labels appear; correct errors and resubmit
- Success message: **"Signup successful! Check your email"**

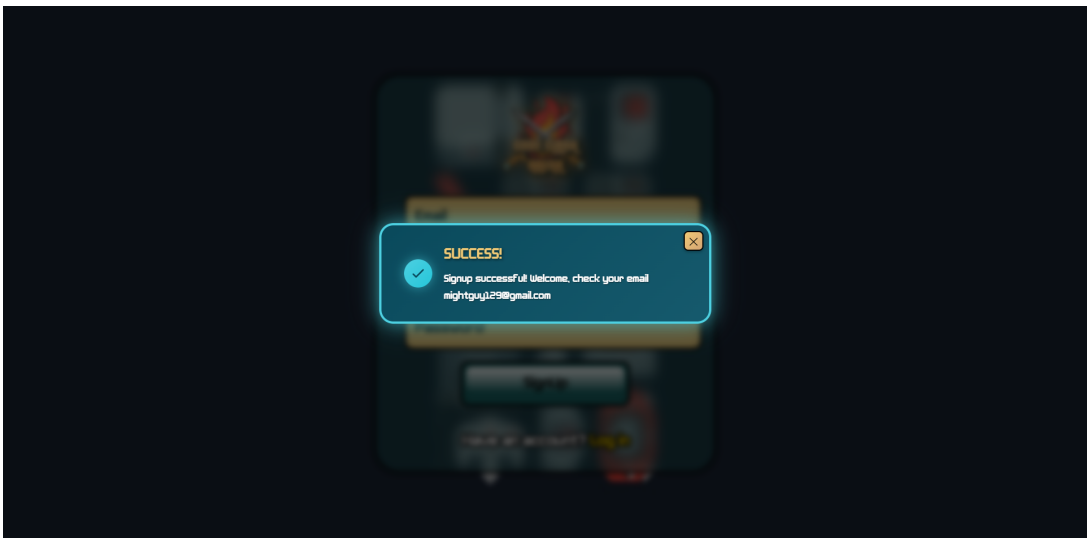


Figure 3.3: Sign-Up Form Completion - Click SignUp button

3.2.4 Step 4: Email Verification

- Check email inbox for verification message from Supabase
- Email may take 1-5 minutes to arrive; check spam folder if needed
- Click **"Verify Email Address"** link in the email
- Your account is now activated

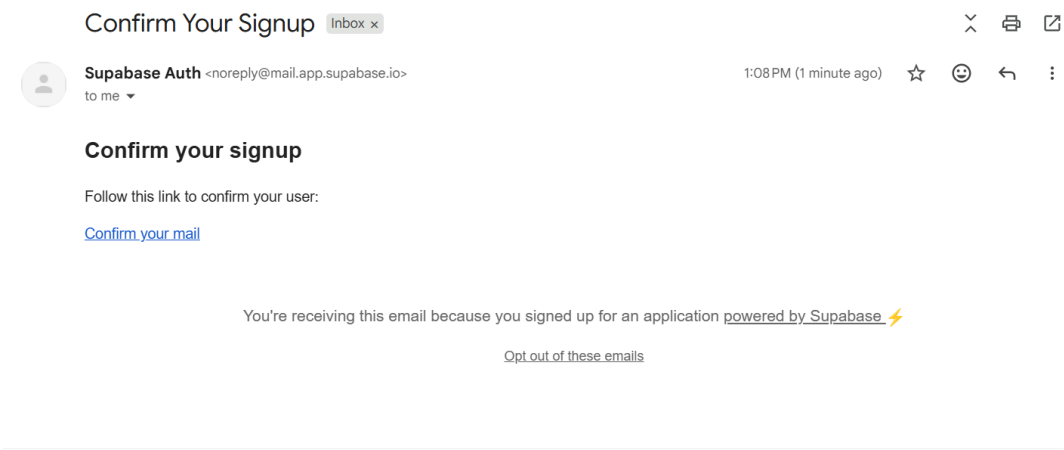


Figure 3.4: Email Verification - Click verification link

3.2.5 Step 5: Login to Complete Setup

- Navigate to Login page and use your registered email and password
- Your Codeforces handle is automatically linked from registration
- Requirements: Existing Codeforces account and verified email before login

3.3 Login Process

3.3.1 Step 1: Navigate to Login Page

- Click **"GET STARTED"** on the homepage
- You are redirected to the Login page

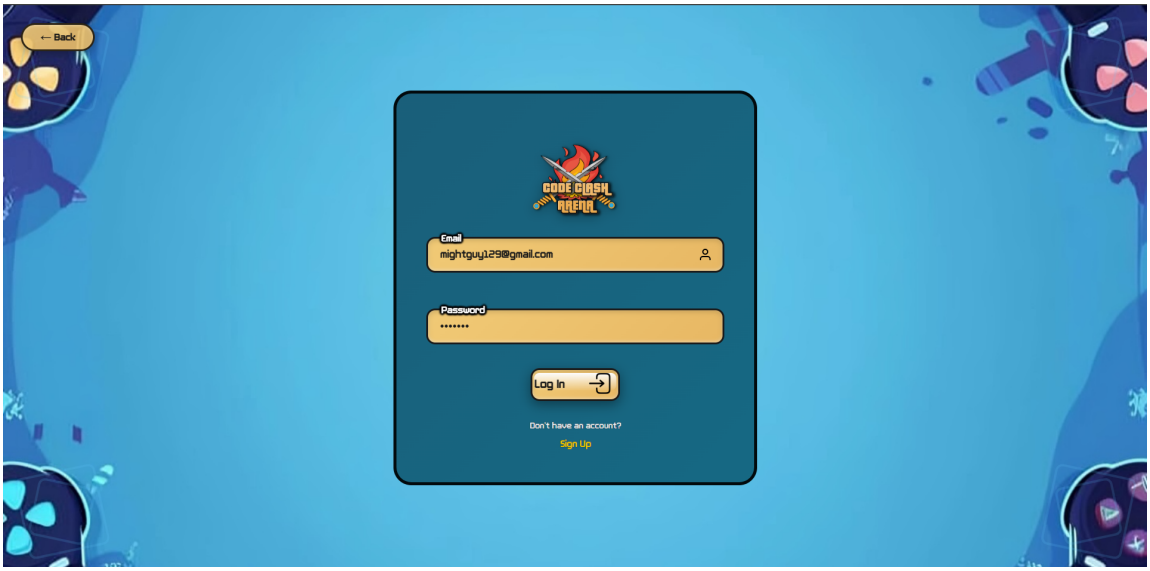


Figure 3.5: Loginpage - Navigate to Login

3.3.2 Step 2: Enter Login Credentials

- Enter your registered email address (case-insensitive)
- Enter your account password (case-sensitive)
- A visibility icon appears in the password field to show/hide password

3.3.3 Step 3: Submit and Authenticate

- Click **"Log In"** button
- System validates credentials with Supabase
- Success message: **"Sign in successful!"**

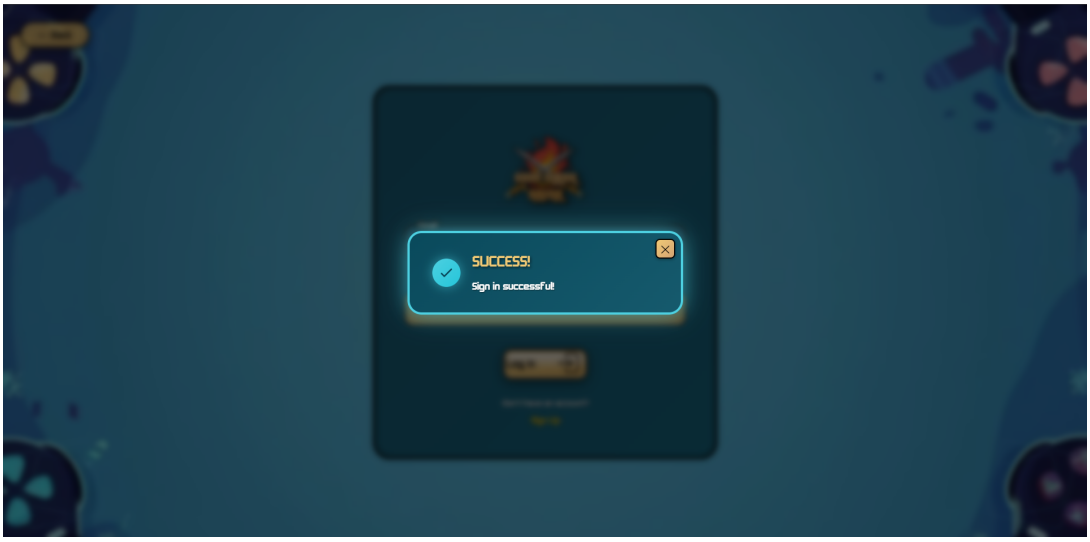


Figure 3.6: Login successful - Success alert displayed

3.3.4 Step 4: Access Dashboard

- After successful login, you are redirected to the Main Dashboard
- Your Codeforces handle is automatically loaded from your account
- All platform features are accessible: PvP Battles, Practice Gym, Clan Wars, and Leaderboards

3.3.5 Troubleshooting

- **Invalid credentials** - Verify email and password are correct; check for typos
- **Email not verified** - Check email inbox for verification link; check spam folder
- **Too many attempts** - Wait 15 minutes before trying again

4 User Roles

CodeClash Arena is designed exclusively for authenticated registered users to ensure fair participation in competitive coding battles. Access to the platform requires account creation, email verification, and a linked Codeforces handle. There are no guest or admin roles in the system - all users have equal access to platform features and compete on a level playing field based on their coding skills and performance.

4.1 Registered User

A registered user is any individual who has successfully completed the sign-up process, verified their email address, and linked a valid Codeforces handle to their account. This is the only role on the platform, and all authenticated users operate with the same base permissions.

4.1.1 Permissions

Registered users have access to the following platform features:

- **Account Management** - Register an account with email and Codeforces handle, log in and log out securely, update profile information and preferences
- **Real-Time 1v1 Coding Battles** - Participate in Local battles (challenge specific friends with custom settings), participate in Global battles (matchmaking with players of similar skill level), compete in time-based coding challenges with live synchronization
- **Clan System** - Create a new clan or join existing clans, participate in clan battles when selected by clan leader, contribute to clan rankings and statistics, manage clan memberships (leaders can approve/reject join requests)
- **Code Submission** - Submit code solutions in multiple programming languages (C++, Java, Python, JavaScript, etc.), receive automated verdicts from Codeforces judge (Accepted, Wrong Answer, Time Limit Exceeded, etc.), test code locally using integrated code editor
- **Statistics and Rankings** - View personal battle history and performance metrics, track XP, level, and ELO-based rating progression, access global and clan leaderboards, view detailed submission verdicts and problem-solving statistics
- **Practice Mode** - Access Practice Gym with thousands of Codeforces problems, filter problems by difficulty, tags, and rating range, receive AI-based logical hints for approach guidance (without revealing full solutions), solve problems without time pressure for skill improvement
- **Social Features** - Send and accept friend requests, view friend profiles and statistics, receive notifications for clan join requests and battle invitations

4.1.2 Restrictions

Registered users are subject to the following restrictions to maintain platform integrity and fair competition:

- **No System Modifications** - Users cannot modify system configurations, settings, or platform code
- **No Ranking Manipulation** - Direct manipulation of ratings, XP, leaderboards, or battle outcomes is strictly prohibited
- **No Private Data Access** - Users cannot access other users' private information, submission code, or system database records
- **No Unauthorized Authentication** - Users cannot share accounts or use multiple accounts to gain unfair advantages
- **Fair Play Requirements** - Code submissions must be original work; plagiarism or copying solutions is not permitted

5 Navigation Guide

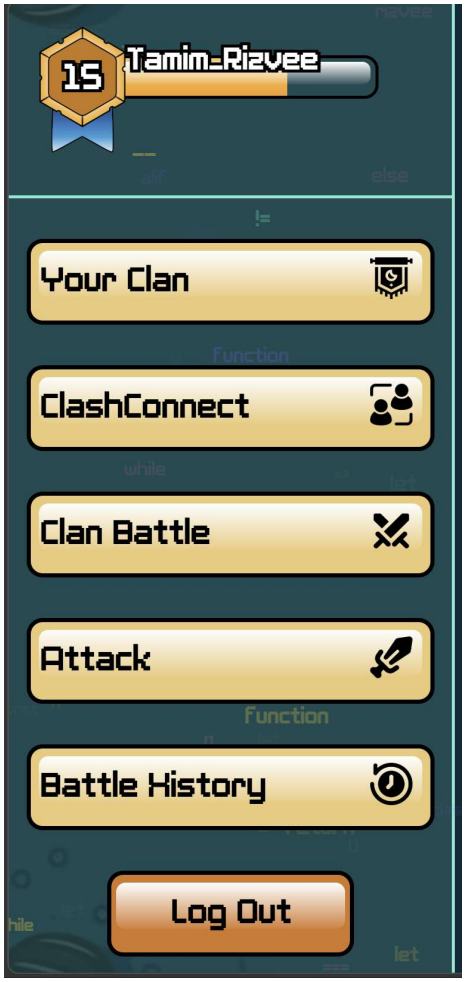
5.1 Menu Structure

After login, the user lands on the main menu dashboard of the game. At the top of the screen, the user's profile information is displayed, showing the player name, the current level, and a progress bar indicating the player's advancement toward the next level.

Below the profile section, the interface presents a vertically arranged menu layout consisting of the following navigation options: **Your Clan**, **ClashConnect**, **Clan Battle**, **Attack**, and **Battle History**. These options are displayed as large, clearly separated buttons to allow quick navigation through the main features of the system.

At the bottom of the menu, there is a **Log Out** button which allows the user to exit the account and return to the login screen. The overall structure provides a simple and organized navigation system for accessing the main functionalities of the application.

> **Note:** The **Card Interface** at the middle of the screen displays what exactly each button does upon hovering them.



(a) Main Menu Buttons

(b) Card Interface

Figure 5.1: Two Interfaces in Landing Page

5.2 Navigation Options

5.2.1 Your Clan

This button launches a floating menu showing the details of current clan the user belongs and details and ratings of the clan members if the current user is in a clan.Otherwise, the floating menu will show two options for finding a clan or creating a clan.

5.2.2 Clash Connect

This button opens a floating menu showing two options. First, it shows a page for finding people and send friend request to them.User can cancel an unaccepted friend request from here. Second, there is a mail page where friend requests sent to user are shown. This menu also shows upcoming clan requests if the user is a leader of a clan.

5.2.3 Clan Battle

This button opens a page where user can participate in (if the leader has chosen current user for battle) or spectate (if the leader has not chosen current user for battle) an ongoing clan battle.

Note: The user must be a member of a clan to avail this feature.

5.2.4 Attack

Launch the PvP battle menu to compete in 1v1 coding challenges. This button opens an overlay menu with multiple battle mode options: Global Battle Mode (matchmaking with players of similar rating), Local Battle Mode (challenge specific friends), and Practice Gym (solo practice without competition). Select your preferred mode to begin competitive coding.

Note: The user must make some friends to participate in the **Local Battle Mode**

5.2.5 Battle History

View your complete battle record and performance statistics. This section displays all your past 1v1 battles and clan battles, showing opponents faced, problems solved, battle outcomes (win/loss), time taken, and rating changes. Use this feature to analyze your performance trends and track your improvement over time.

5.2.6 Log Out

Securely end your current session and return to the homepage. Clicking this button clears your authentication tokens and logs you out of the platform. Always use this option when accessing CodeClash Arena from shared or public computers to protect your account security.

5.3 Buttons and Icons

Button / Icon	Function
Close Button	Closes an overlay menu.
Leave Button	Enables user to leave a clan.
Level Icon	Clicking on the level copies users' unique ID to the clipboard.
Accept Button	Enables user to accept clan or friend requests and local or global battle requests.
Reject Button	Enables user to reject clan or friend requests and local or global battle requests.
Challenge Button	Enables user to send a 1v1 challenge to a friend.
Send Request Button	Enables user to send friend request to an existing user or clan request to an existing clan.
Cancel Request Button	Enables user to cancel friend request to an existing user or clan request to an existing clan.
Enter Button	Enables user to enter 1v1 battle mode or practice gym.
Exit Button	Enables user to exit from 1v1 battle mode or practice gym.
Submit Button	Enables user to submit a problem to Codeforces.
Search Button	Start the search according for caln or a existing user.
View Detail button	Hovering this button shows friends' details.

Table 5.1: Functions of Different Buttons and Icons

6 Error Handling and Messages

6.1 Authentication Errors

Error Message	Possible Cause	Solutions
"Invalid login credentials"	The entered email or password is incorrect, or formatting issues like extra spaces or wrong capitalization exist.	• Check email and password • Ensure Caps Lock is off • Remove extra spaces • Use email instead of Codeforces handle • Verify email after signup
"Email not confirmed" or "Error fetching user data"	The account has not been verified via email, preventing successful login.	• Check inbox or spam for verification email • Click confirmation link • Retry login after verification
"Too many login attempts"	Multiple failed login attempts have temporarily locked the account for security reasons.	• Wait 15–30 minutes before retrying
"User already registered" or "Email already exists"	The email is already associated with an existing account.	• Use login page • Reset password if forgotten • Use another email for new account
Field validation error or "Invalid handle"	The Codeforces handle provided is incorrect or does not exist.	• Verify handle on Codeforces profile page • Check spelling and capitalization • Create account if needed

Table 6.1: Common Login and Authentication Errors

6.2 Network and Connectivity Errors

Error Message	Possible Cause	Solutions
"Network error" or "Failed to fetch"	The browser or application could not connect to the server due to internet issues, VPN restrictions, firewall blocks, or cached data problems.	• Check internet connection • Refresh page (F5) • Verify firewall settings • Disable VPN temporarily • Clear browser cache • Try a different browser
"Request timeout" or "Connection timeout"	The server did not respond in time, often caused by slow or unstable internet, high bandwidth usage, or network congestion.	• Ensure stable connection (5+ Mbps for battles) • Close bandwidth-consuming applications • Switch to a wired connection • Try again after a moment

Table 6.2: Common Network and Timeout Errors

6.3 Clan Related Errors

Error Message	Possible Cause	Solutions
"Already in a clan" or "Join request failed"	The user is currently a member of another clan, or the target clan is not accepting new members or has pending requests.	• Leave current clan before joining another • Wait for pending requests to be processed • Verify clan is accepting members and you meet requirements
"Insufficient permissions" or "Only clan leader can start battles"	The user does not have the required leadership permissions to initiate a clan battle.	• Contact clan leader to start battle • Create your own clan for leadership permissions

Table 6.3: Common Clan-related Errors

6.4 Battle Related Errors

Error Message	Possible Cause	Solutions
"Connection lost" or "Battle disconnected"	The connection between the client and server was interrupted, often due to unstable internet or network congestion. Failure to reconnect may result in automatic forfeit.	• Use wired Ethernet connection • Ensure stable internet before battles • Close unnecessary applications • Avoid battles during network instability
"Submission failed" or "Error submitting code"	The code submission could not reach the server or was rejected due to incorrect configuration, network issues, or account problems.	• Check internet connection • Verify Codeforces handle is correctly linked • Ensure Codeforces account is in good standing • Retry after a few seconds • Verify correct programming language is selected

Table 6.4: Common Battle and Submission Errors

Note: Users must not open any screen recorder, snipping tool, or use third-party applications during code submission. Doing so may prevent the submission from being processed, and the problem may not be fetched and displayed. Codeforces has increased its security to ensure fairness and transparency in competitive programming.